

# Represent natural numbers in expanded notation using place value and powers of 10

## Learning intention

- represent natural numbers in expanded notation using place value and powers of 10.

## Teach and model

- relating the sequences 10, 100, 1000, 10 000 and  $10^1$ ,  $10^2$ ,  $10^3$ ,  $10^4$ .
- applying and explaining the connections between place value and expanded notations.
- investigating exponent notation for powers of 10 such as “one hundred thousand” is  $100\,000 = 10 \times 10 \times 10 \times 10 \times 10 = 10^5$ .

## Check understanding

- Explain the main idea and give one accurate example.
- Show the idea in a second way: with words, a model, a diagram or symbols.
- Apply the idea to a short unfamiliar situation and justify the result.

## Teaching cautions

- Ask the student to explain their choice, not only state an answer.
- Check that key vocabulary is understood in a new example.
- Mix representations so the skill is transferred rather than copied.

## Quick lesson sequence

- Connect to prior knowledge and introduce the key vocabulary.
- Model one clear example, then solve a second example together.
- Finish with an independent check and a short student explanation.